

Matteo Calviello

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SUMMARY

Associate Data Scientist at Exelon (ComEd) building computer vision, machine learning, and AI agent solutions for utility infrastructure. Computer Science and Data Science graduate from Rose-Hulman Institute of Technology.

EDUCATION

Bachelor of Science, Computer Science and Data Science & AI (Double Major)	GPA: 3.86
Rose-Hulman Institute of Technology	May 2026
<i>Honors:</i> Magna Cum Laude; Michael Atkins Best Senior Thesis Award; Frank Young Outstanding Scholar Award	

EXPERIENCE

Associate Data Scientist - Exelon ComEd - Chicago, IL	August 2026 - Present
– Engineering AI pipelines within the Grid Analytics department, applying computer vision and machine learning to optimize utility infrastructure. – Developing a company-wide chatbot utilizing AI agents to automate data retrieval and streamline internal processes. – Executing complex data manipulation and preprocessing using SQL to feed structured datasets into AI models.	
Data Science Intern - Exelon ComEd - Chicago, IL	Summer 2023, Summer 2024, Summer 2025
– Built CV pipeline to predict pole accessibility and replacement costs (94% SME agreement); integrated model outputs into cost optimization algorithm, reducing budgeting error by 90%. – Developed ViT and CNN models for pole component detection and pole base identification (up to 86% accuracy), improving automated field identification. – Automated image labeling with clustering, vector embeddings, and Meta’s Segment Anything Model (SAM), achieving a 10x speedup across 900k+ pole images. – Adapted radar analysis scripts for storm detection and infrastructure monitoring.	

PROJECTS

Single Image Super-Resolution Research (Best Senior Thesis Award) - Github	May 2025 – May 2026
– Reproduced Omni Aggregation Network (OAN) and validated PSNR/SSIM on standard benchmarks within $\pm 5\%$. – Improved reconstruction quality and reduced parameters by 10% via architectural enhancements. – Developed full PyTorch training and evaluation pipeline with custom data loaders for efficient experimentation.	
2026 NCAA Analytics Challenge - Github	
1st Place Overall (157 Teams)	February 2026 - April 2026
– Built predictive models using NCAA basketball data to identify championship success factors. – Created custom team performance metrics and upset prediction models for tournament analysis. – Developed Tableau dashboards to present tournament insights and business recommendations to NCAA judges.	
NeuroScan AI - Github	
Medtronic Top 3 Finalist	May 2025 - Nov 2025
– Developed a full-stack medical analysis platform with a dynamic frontend for AI-powered PDF report generation. – Integrated three TensorFlow models (U-Net, CNNs) for brain scan segmentation and tumor classification. – Top 3 finalist, 2025 BMES Student Design Competition; presented at BMES conference in Oct 2025.	
Lunar Autonomy Challenge	
NASA Top 8 Finish	November 2024 - May 2025
– Built stereo vision and perception system to enable hazard detection in simulated lunar navigation. – Engineered algorithms and ML models to process lunar data, generate elevation maps, and identify obstacles.	

SKILLS

Languages:	Python, SQL, R, Java, C, JavaScript, MATLAB
AI / Data Science:	Machine Learning, Deep Learning, Computer Vision, Agentic AI, Predictive Modeling, Data Analysis, Data Visualization
Frameworks & Tools:	PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, MLflow, Tableau, Power BI, GeoPandas
Cloud & DevOps:	Docker, Kubernetes, Azure, CI/CD, Git, Linux, Podman, Containerization

ACTIVITIES

– Alpha Tau Omega Fraternity, Executive Board Member – Vice President, Recruitment Chair	2022 - 2026
– Teaching Assistant – Database Systems, Artificial Intelligence, Parallel Computing, Machine Learning	2023 - 2026
– Rose-Hulman DIII Varsity Swimming & Diving Captain, 4x HCAC Conference Champion	2022 - 2026